

Client Change Request (CCR)**CCR Number:** CCR-004**Client:** Precision Payroll Services**Submitted By:** Amanda Carter, Payroll Supervisor**Priority:** High**Requested Delivery:** End of Today's Sprint**Assigned Developer:** _____**Date Assigned:** _____

Business Background

Precision Payroll Services processes employee payroll information for several small businesses. At the end of each week, employee work hours are stored in a text file that is received from another department. Before payroll is processed, the file must be reviewed by a supervisor.

Currently, supervisors manually number each line in the report to make it easier to reference individual employee records during payroll verification. This process is time-consuming and increases the chance of mistakes.

The payroll department would like a simple computer program that automatically creates a numbered copy of the original file while providing a summary of the file that was processed.

At this time, the payroll department only wants to process one input file at a time.

Customer Requirements

The employee should be able to specify the name of a text file containing employee work hours.

The program should:

- Open the specified input file.
- Create a new output file.
- Copy every line from the input file into the output file.
- Number each line in the output file so that supervisors can easily reference specific records.
- Keep track of how many lines were copied.
- Keep track of how many characters were copied from the input file.
- Display a summary after the copy operation has completed.

If the specified input file cannot be opened, the program should display an understandable error message instead of terminating unexpectedly.

The output should be clear, professional, and easy for employees to understand.

Acceptance Criteria

The customer will consider the software complete when all of the following are true.

- ☐ Employee can specify the input file.
- ☐ Program successfully opens an existing text file.
- ☐ Program creates a new output file.
- ☐ Every line from the input file is copied to the output file.
- ☐ Every line in the output file is numbered.
- ☐ Program correctly reports the total number of lines copied.
- ☐ Program correctly reports the total number of characters copied.
- ☐ Program displays an understandable error message if the input file cannot be opened.
- ☐ Program terminates normally after processing the file.

Out of Scope

The customer is **not** requesting:

- Editing employee records
- Calculating employee pay
- Modifying the contents of the input file
- Processing multiple files during one execution
- Sorting employee records
- Saving summary statistics to another file
- Printing reports
- Reading binary files
- Displaying graphical user interfaces (GUI)

If these features appear in the software, they will not be evaluated.

Example Scenario 1

Input File (employee_hours.txt)

John Smith,40

Mary Jones,38

Carlos Garcia,45

Emily Brown,32

Expected Output File (numbered_hours.txt)

1. John Smith,40

2. Mary Jones,38

3. Carlos Garcia,45

4. Emily Brown,32

Program Output

Precision Payroll Services

File Processing Summary

Input File : employee_hours.txt

Output File : numbered_hours.txt

Lines Copied : 4

Characters Copied : 59

File copied successfully.

Example Scenario 2

Employee Input

Input File: payroll.txt

(The file does not exist.)

Expected Output

Precision Payroll Services

File Processing Summary

Error

Unable to open file "payroll.txt".

Please verify that the file exists and try again.

Notes from the Customer

"Our supervisors spend too much time numbering reports by hand. I simply want an exact copy of the original file with line numbers added so everyone can easily discuss individual employee records. If the file cannot be opened, the employee should immediately know what went wrong."

Developer Notes

This section intentionally left blank.

Students will complete the software requirements during class before beginning implementation.

Questions for the Customer

Before writing any code, list any questions you would ask the customer to clarify the request.

- 1.

Reminder

The customer has intentionally described the problem using **business language**, not programming terminology. During class, your first task is to translate these customer requirements into **software requirements** before writing any C code.

For this assignment, you should use only the C language features that have been covered through **Lesson 5**, including:

- File input/output
- Character processing
- Type conversion (casting)
- Basic input/output
- Selection statements
- Loops
- Variables and arrays (as previously learned)

Do **not** use C language features that have not yet been introduced in the course.